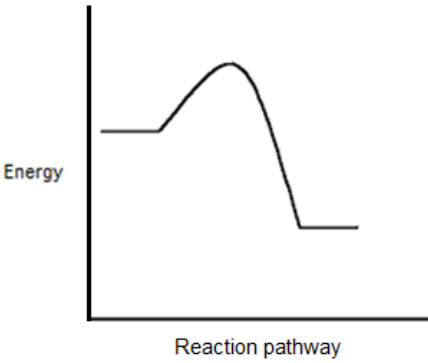


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Agree because we are not told / do not know that the carbonate is in excess (1) there may not be enough to neutralise all of the acid (1) OR Disagree because the carbonate will be in excess (1) so all of the acid will all be used up / neutralised (1) no credit if no choice is made and the answer does not mention agreeing or disagreeing	2			2		2
		(ii)		$\text{K}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{KCl} + \text{CO}_2 + \text{H}_2\text{O}$ reactants (1) products (1) balancing (1) balancing mark can only be awarded if both the reactants and products are correct		3		3	1	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		will overshoot / go past the point of neutralisation / endpoint 'too much acid' is a neutral answer	1			1		1
		(ii)		$H^+ + OH^- \rightarrow H_2O$ (1) charges on ions must be present H^+ comes from the (sulfuric) acid and OH^- comes from the (potassium) hydroxide / alkali (1) 'one comes from the acid and the other from the alkali' and 'they come from the acid and alkali' are neutral answers	2			2		
		(iii)			1			1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		both will give a lilac flame / same colour flame (because both contain the potassium ions / K^+ / potassium)			1	1		1
		(ii)		add silver nitrate solution / $AgNO_3(aq)$ (1) potassium chloride gives a (white) precipitate whereas potassium sulfate gives no precipitate / only the chloride gives a (white) precipitate (1) OR add barium chloride solution / $BaCl_2(aq)$ (1) potassium sulfate gives a (white) precipitate whereas potassium chloride gives no precipitate / only the sulfate gives a (white) precipitate (1)		1	1	2		2
				Question 4 total	6	4	2	12	1	6